

AGROS – Cardamom Estate Management System

Database Design

This document describes the database design for the AGROS – Cardamom Estate Management System. The purpose of this database is to store, manage, and retrieve information related to estate management, crop activities, labor management, and production records in a centralized and efficient manner.

1. Users Table

This table stores authentication and role-based access details for Admin, Estate Manager, and Worker.

Field Name	Data Type	Description
user_id	INT (PK)	Unique user identifier
username	VARCHAR(50)	Login username
password	VARCHAR(255)	Encrypted password
role	ENUM	User role (Admin, Manager, Worker)
status	ENUM	Account status

2. Estate Table

Stores basic information related to the cardamom estate.

Field Name	Data Type	Description
estate_id	INT (PK)	Estate identifier
estate_name	VARCHAR(100)	Name of the estate
location	VARCHAR(150)	Estate location
total_area	DECIMAL	Total estate area

3. Plot Table

Each estate is divided into multiple plots for effective management.

Field Name	Data Type	Description
plot_id	INT (PK)	Plot identifier
estate_id	INT (FK)	Linked estate
plot_name	VARCHAR(50)	Plot name
plant_count	INT	Number of cardamom plants

4. Activity Category Table

Defines different types of agricultural activities performed in the estate.

Field Name	Data Type	Description
category_id	INT (PK)	Category identifier
category_name	VARCHAR(50)	Activity name
description	TEXT	Activity details

5. Production Table

Stores yield and production details after harvesting.

Field Name	Data Type	Description
production_id	INT (PK)	Production identifier
plot_id	INT (FK)	Associated plot
harvest_date	DATE	Date of harvest
quantity	DECIMAL	Yield quantity